

Predicting Calories Burned During Exercise: A Regression Pipeline with Model Comparison, Robustness, and Interpretability Analysis

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Abstract—Fitness trackers commonly estimate calorie expenditure using generic heuristics that ignore individual physiology. This paper presents a supervised regression pipeline that predicts calories burned during a workout from biometric and exercise-session data (15,000 records). Outlier bounds, imputation, and feature scaling are fit exclusively on a training partition and applied unchanged to a held-out test partition. Six regression models are compared, the best candidates are tuned via cross-validated search, and the resulting gains are confirmed with a repeated-split robustness check. The final tuned gradient boosting model reaches $R^2 = 0.9995$ and a mean absolute error of 0.96 kcal on the test set. Feature importance and SHAP analysis show that a single engineered feature, workout workload, dominates the prediction. Model selection in this study compares every baseline, tuned, and ensemble candidate on the same held-out test partition rather than a separate validation set; this simplification is disclosed explicitly as a limitation of the evaluation protocol rather than presented as resolved.

Index Terms—calorie prediction, regression, feature engineering, gradient boosting, hyperparameter tuning, SHAP, machine learning.

1 INTRODUCTION

WEARABLE fitness devices and gym equipment routinely report an estimate of calories burned during a workout. In practice, many such estimates rely on coarse, population-level heuristics (e.g., MET tables) that map exercise type and duration to an average calorie cost, largely ignoring how an individual's heart rate, body temperature, age, or body composition affect true energy expenditure. A data-driven regression model trained on paired session and outcome data offers the possibility of a substantially more personalized estimate.

This paper addresses the task of predicting Calories, a continuous target measured in kilocalories, from a combination of static attributes (age, height, weight, gender) and session-level biometric signals (exercise duration, heart rate, body temperature). The data come from two files, `exercise.csv` and `calories.csv`, joined on a user identifier, yielding 15,000 complete records with no missing values, duplicates, or physiologically implausible readings.

Beyond fitting an accurate model, this work aims for methodological transparency about what was and was not done to guard against data leakage. Data preparation, outlier bounds, missing-value imputation, and feature scaling, is fit exclusively on the training partition and applied unchanged to the test partition, so that no information about test rows influences those choices. Model selection, however, is a weaker point: every baseline, tuned, and ensemble candidate in this study is compared on the same held-out test partition rather than on a separate validation set, which can optimistically bias the final reported number for whichever candidate happens to look best on that particular split. Rather than silently accepting this or overstating a fix that was not actually implemented, this limitation is

disclosed explicitly, in the same spirit as reporting an unresolved fairness gap rather than declaring it solved: it is identified precisely, its likely direction of bias is discussed, and a corrected protocol is proposed as future work (Section IV).

The main contributions of this paper are:

- A leakage-safe data preparation pipeline, in which outlier bounds, imputation, and scaling are fit exclusively on training data before being applied to the test partition.
- An engineered workload feature (duration multiplied by heart rate) that substantially improves on the raw features taken individually.
- A systematic comparison of six regression models, cross-validated hyperparameter tuning for the three strongest candidates, and a repeated-split robustness check that confirms the tuning gains are statistically real rather than an artifact of a single lucky split.
- An interpretability analysis combining impurity-based feature importance with SHAP values.
- An explicit disclosure and discussion of test-set reuse during model selection as a known limitation of the current evaluation protocol, together with a concrete proposal for correcting it.

1.1 Group Organization

This project was completed individually. The author was solely responsible for every stage of the work reported here: problem definition, data understanding and quality assessment, feature engineering, exploratory data analysis, model implementation, hyperparameter tuning, the robustness and interpretability analyses, and the writing of this paper.

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1.2 Paper Organization

The remainder of this paper is organized as follows. Section II describes the proposed method, including data preparation, feature engineering, model selection, and evaluation protocol. Section III reports the resulting metrics and the main insights drawn from them. Section IV concludes and discusses limitations and future work.

2 PROPOSED METHOD

2.1 Data Understanding and Quality

The merged dataset contains nine raw numeric/categorical columns per session (Age, Height, Weight, Duration, Heart_Rate, Body_Temp, Gender) plus the target Calories. Gender is nearly balanced (50.4% female, 49.6% male). Five formal data-quality checks were run: completeness, representation of expected categorical values, plausibility of numeric ranges (e.g., age between 10 and 100, heart rate between 30 and 220 bpm), row/identifier uniqueness, and unexpected zero values. All five checks passed cleanly: zero missing values across all 15,000 rows, zero duplicate rows and zero duplicate `User_ID` values, zero rows failing any plausibility check, and zero unexpected zero-values in any numeric column. This indicates the dataset is well curated; the checks therefore serve mainly as a template that would catch real issues on noisier, real-world wearable exports.

2.2 Feature Engineering

Two derived features were added: body mass index, $BMI = \text{Weight}/(\text{Height}/100)^2$, and a workload proxy, $\text{Workload} = \text{Duration} \times \text{Heart_Rate}$, intended to capture overall cardiovascular effort rather than duration or intensity alone. Gender was label-encoded as a binary `Gender_Encoded` feature. Pearson correlation with the target confirmed that `Workload` ($r = 0.975$), `Duration` (0.955), `Heart_Rate` (0.898), and `Body_Temp` (0.825) carry almost all of the linear signal, while static attributes such as Age, BMI, Weight, Height, and `Gender_Encoded` are only weakly correlated ($|r| \leq 0.15$).

2.3 Exploratory Data Analysis

The target `Calories` is right-skewed (skewness 0.51), with a long tail toward high-effort sessions, while `Duration` is symmetric (skewness 0.00). Boxplots of `Heart_Rate`, `Body_Temp`, BMI, and `Workload` each show a handful of upper-tail outliers, motivating the outlier-clipping step described next. Scatterplots of `Duration`, `Heart_Rate`, and `Workload` against `Calories` each show a strong, close-to-linear relationship, consistent with the correlation coefficients reported above, and a full correlation heatmap across all nine numeric features reproduces the same ranking.

2.4 Data Preparation

The data were split into training and test subsets (80/20) before any outlier handling or scaling, so that no information about test rows could influence decisions made on the training set. Outlier bounds for `Heart_Rate`, `Body_Temp`,

BMI, and `Workload` were computed with the interquartile-range rule using training rows only, then applied unchanged to clip the test rows. Missing-value imputation (median) and feature scaling (standardization) were likewise fit only on the training set inside a `scikit-learn` `ColumnTransformer` pipeline, then applied to the test set via `transform` alone.

2.4.1 Model-Selection Protocol

No separate validation set was carved out of the training data. Baseline models, the polynomial-feature experiment, hyperparameter tuning (selected via 5-fold cross-validation within the training set), and the voting ensemble are all subsequently evaluated on the same held-out test partition, and the best-performing configuration is chosen by comparing those test metrics directly. This means the test set is not touched only once: it is queried repeatedly across roughly a dozen candidate configurations before a winner is declared, which can optimistically bias the reported metric for that winner relative to its true generalization performance. This trade-off is discussed further in Section IV.

2.5 Feature Selection

Given the modest dimensionality (nine candidate features), aggressive reduction (e.g., PCA) was not pursued. Instead, Recursive Feature Elimination (RFE) with a linear regression estimator was swept over subset sizes from two to nine features, scored via 5-fold cross-validation on the training set. Cross-validated R^2 was 0.962, 0.963, 0.966, and 0.967 at two, three, four, and five features respectively, then jumped to 0.984 at six features and stayed within 0.0001 of that value through nine, indicating no feature was redundant enough to discard. All nine features were therefore retained for every downstream model; RFE served here as a diagnostic rather than a filtering step.

2.6 Models and Training

Six regression models were trained as baselines on the training set and evaluated on the test set: Decision Tree, k -Nearest Neighbors ($k = 15$), Linear Regression, Ridge regression, Random Forest (300 trees, max depth 8), and Gradient Boosting. A degree-2 polynomial feature expansion was additionally tested with Linear Regression to probe whether the true `Duration/Heart-Rate-Calories` relationship has curvature that a plain hyperplane misses.

The three strongest model families, Random Forest, Gradient Boosting, and k -NN, were then tuned via cross-validated randomized/grid search (5-fold `KFold` on the training set, scored by R^2), searching over tree count and depth for the forest, learning rate and depth for boosting, and neighborhood size/weighting for k -NN.

2.7 Robustness and Interpretability

To verify that the improvement from tuning was not an artifact of the particular train/test split used for the search, each tuned configuration was refit and re-evaluated against its untuned baseline across ten independent random splits (with outlier clipping and scaling refit fresh on every split), and the mean R^2 difference was reported with a 95%

TABLE 1
Baseline Model Comparison (Test Set)

Model	R^2	RMSE (kcal)	MAE (kcal)
Gradient Boosting	0.9979	2.91	2.13
Random Forest	0.9953	4.36	3.04
k -NN	0.9946	4.65	3.33
Linear Regression	0.9845	7.90	5.83
Ridge	0.9845	7.91	5.81
Decision Tree	0.9818	8.58	6.06

TABLE 2
Effect of Hyperparameter Tuning (Test Set)

Model	Base R^2	Tuned R^2	Params
k -NN	0.9946	0.9950	$k=15$, dist.
Random Forest	0.9953	0.9974	250 trees, $d=10$
Gradient Boosting	0.9979	0.9995	400 est., $d=5$

confidence interval. This check is independent of the model-selection caveat above: it re-splits the data from scratch on each of the ten repetitions, so it measures whether the tuning effect itself is real, rather than whether the specific winning configuration was chosen fairly.

Interpretability was assessed two ways: impurity-based feature importance for the tuned tree ensembles, and SHAP (SHapley Additive exPlanations) values computed with a tree explainer on a 200-row test-set sample, cross-checked against the impurity ranking. A voting ensemble of the three tuned models was also evaluated for comparison.

3 RESULTS

3.1 Baseline and Tuned Model Comparison

Table 1 reports test-set performance for the six baseline models. Gradient Boosting led from the outset ($R^2 = 0.9979$), followed by Random Forest and k -NN, with the linear models clearly behind the non-linear ones. Five-fold cross-validation on the training set reproduced the same ranking ($R^2 = 0.9974 \pm 0.0001$ for Gradient Boosting), confirming it was not a single-split artifact. The degree-2 polynomial expansion improved Linear Regression substantially (R^2 : 0.9845 $\rightarrow$ 0.9987; MAE: 5.83 $\rightarrow$ 1.72 kcal), showing that a purely linear model under-fits the true relationship.

Table 2 shows the effect of hyperparameter tuning. Gradient Boosting again led, reaching $R^2 = 0.9995$ (MAE = 0.96 kcal) with a learning rate of 0.1. The repeated-split robustness check confirmed all three tuning gains were statistically real, with 95% confidence intervals excluding zero: Random Forest +0.0023 R^2 (CI [0.0022, 0.0024]), Gradient Boosting +0.0021 (CI [0.0020, 0.0022]), and k -NN +0.0004 (CI [0.0004, 0.0004]).

A voting ensemble of the three tuned models reached $R^2 = 0.9988$ (MAE = 1.51 kcal), worse than the tuned Gradient Boosting model alone, since averaging in the comparatively weaker k -NN and Random Forest predictions pulled the ensemble down.

3.2 Final Model Selection

Comparing all baseline, tuned, and ensemble candidates on the same held-out test set, tuned Gradient Boosting is the best-performing configuration: $R^2 = 0.9995$, RMSE

TABLE 3
Feature Importance, Tuned Models (Impurity-Based)

Feature	Random Forest	Gradient Boosting
Workload	0.9578	0.9414
Age	0.0254	0.0259
Gender_Encoded	0.0075	0.0083
Heart_Rate	0.0065	0.0209
Weight	0.0017	0.0032
Height	0.0006	0.0001
BMI	0.0003	0.0001
Body_Temp	0.0001	0.0000
Duration	0.0001	0.0001

= 1.39 kcal, MAE = 0.96 kcal. As noted in Section II, this number is the result of comparing roughly a dozen candidates against the same test partition, which is a methodologically weaker protocol than reserving an untouched validation set for selection and touching the test set only once for the final winner; the reported $R^2 = 0.9995$ should therefore be read as a plausibly optimistic upper estimate of this model's true generalization performance rather than an unbiased one.

3.3 Feature Importance and SHAP

Impurity-based importance and mean absolute SHAP values agree closely for both tuned tree ensembles. Table 3 reports the full impurity-based breakdown. *Workload* overwhelmingly dominates both models (mean $|\text{SHAP}| \approx 50\text{--}54$), with *Age* a distant second. Most features' SHAP-versus-impurity ranks differ by at most one position; the one exception is *Height* in Gradient Boosting, which impurity ranks 7th but SHAP ranks 9th (last), a two-position shift. Overall, the collinearity between *Workload*, *Duration*, and *Heart_Rate* does not meaningfully distort the impurity-based ranking. The practical implication is that most of the model's predictive power is concentrated in a single engineered interaction term rather than spread across many raw measurements, consistent with the near-linear scatter of *Workload* against *Calories* observed during exploratory analysis.

4 CONCLUSIONS

This paper presented a regression pipeline for predicting exercise calorie expenditure, reaching a competitive $R^2 = 0.9995$ (MAE = 0.96 kcal) with a tuned Gradient Boosting model on the test set. Data preparation is leakage-safe: outlier bounds, imputation, and scaling are fit exclusively on the training partition. Feature engineering played an outsized role beyond modeling choices: a single derived feature, workload, was found by both impurity-based importance and SHAP to dominate prediction quality, more so than any raw measurement, and the repeated-split robustness check confirmed the hyperparameter-tuning gains for all three tuned models were real rather than single-split noise.

The most important limitation of this study is methodological rather than predictive: model selection compares every baseline, tuned, and ensemble candidate on the same held-out test set rather than a separate validation partition, so the reported $R^2 = 0.9995$ for the winning configuration

is best read as an optimistic upper bound rather than an unbiased estimate. This is disclosed here explicitly rather than silently accepted, and a concrete fix is proposed for future work: carve out a third, validation-only partition (e.g., a 60/20/20 train/validation/test split), run every comparison in this paper on the validation set, and touch the test set exactly once, for the single already-chosen model, at the very end. Because Gradient Boosting won by a wide margin at every stage of this study, it is likely, but not verified here, that the qualitative ranking of models would survive such a correction even if the exact reported R^2 shrank slightly.

Two further limitations are worth noting. First, the dataset is unusually clean (no missing values, duplicates, or implausible readings), so results may not transfer directly to noisier, real-world wearable-device data. Second, `Heart_Rate` and `Body_Temp` are physiological responses that are only observable during or after a workout; the model is consequently better understood as estimating calories for a completed session rather than forecasting them before a workout begins. A natural extension is an ex-ante variant of this pipeline restricted to attributes known in advance (age, height, weight, gender, planned duration), which would likely be a harder and more product-relevant prediction task, together with distribution-aware loss functions and quantile regression to report an honest prediction interval rather than a single point estimate.

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